

PORTABLE 15-WATT PA SYSTEM

With Battery Management

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Many mini electronic modules are available in the market for providing quick plug-and-play solutions. We can use some of these modules to make a low-cost, portable, 15W public address (PA) system that includes a battery management system (BMS).

Construction and working

Fig. 1 shows connection diagram of the project, which uses three main modules along with three lithium-ion batteries (B1, B2, and B3). There are following main blocks in this project:

1. LM386 amplifier module, which acts as preamplifier
2. TDA2030 audio amplifier module, which acts as a power amplifier
3. 3S 10A, 12V 18650 lithium-ion battery charger protection module
4. Three 3.7V 18650 lithium-ion battery banks
5. 12V DC power supply

LM386 audio amplifier module. It is a miniature module with an on-board LM386 chip (700mW analogue input, class-AB audio amplifier). It is a simple operational amplifier (op-amp), which is best suited for low-power audio amplification applications. Fig. 2 shows the amplifier module, which has a gain of around 200. It has an on-board 10-kilo-ohm variable resistor to adjust the gain. There is an on-board power indicator LED too.

The module has a generic operating voltage between 4 and 12V DC and an on-board potentiometer to increase/decrease the volume. It has low static power consumption and quiescent current of about 2mA that is suitable for a battery-powered

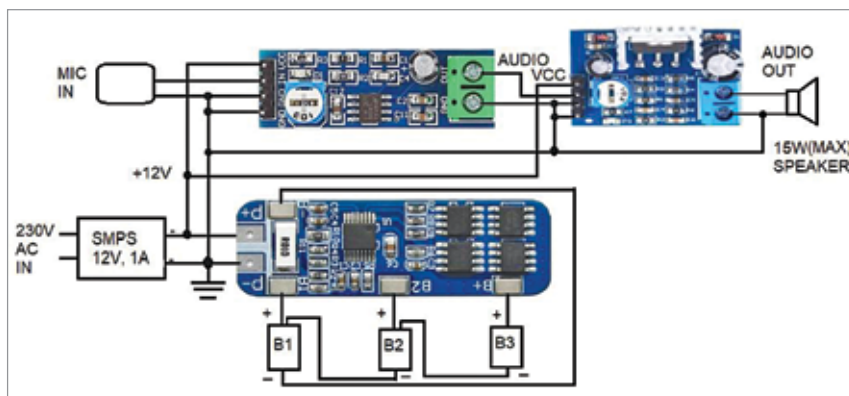


Fig. 1: Connection diagram for the PA system



Fig. 2: LM386 audio amplifier module

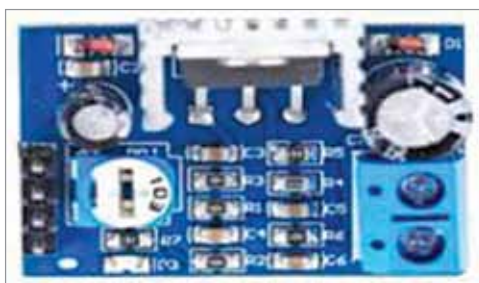


Fig. 3: TDA2030A audio amplifier module

device. It has four pins, namely, VCC, Vin, audio GND, and supply GND. Due to its high gain and low distortion, the board can be used as a preamplifier. This helps to connect the module easily to a microphone.

A dynamic microphone is used here, which acts as an input to the preamplifier module. A dynamic

microphone is a transducer audio device that produces sound through a moving electrically conductive diaphragm and a permanent magnetic structure. Dynamic microphone has simple advantages like plug and play, output electrical signal that is directly proportional to the input voice, and electrical voltage output of the order of a few millivolts. Whereas a condenser microphone needs a little extra circuitry to drive the preamplifier module. The output of the preamplifier module is connected to TDA2030A audio amplifier module (Fig. 3).

TDA2030A audio amplifier module. This module is based on the famous TDA2030A audio amplifier IC and provides 18W power output with very low harmonic distortion. It has an operating voltage range of 6 to 12V with 10-kilo-ohm adjustable resistance to adjust the volume and an onboard LED power indicator. A 15W speaker can be used to connect to the